pk2cmd Usage Guide

Complete guide for using pk2cmd with PICkit2/PICkit3 programmers on macOS.

PICkit2/PICkit3 Hardware Setup

PICkit2 Pinout (6-pin ICSP Connector)

```
PICkit2 Connector (looking at programmer end):
┌─────────────────────┐
│ 1   2   3   4   5   6 │
└─────────────────────┘

Pin 1: VPP/MCLR  (Programming voltage / Master Clear)
Pin 2: VDD       (Target power supply +3.3V or +5V)
Pin 3: VSS       (Ground)
Pin 4: PGD       (Programming Data)
Pin 5: PGC       (Programming Clock)
Pin 6: AUX       (Auxiliary / not used for basic programming)
```

Standard ICSP Connection to Target

```
Target PIC          PICkit2 Pin     Wire Color (typical)
────────────        ───────────     ──────────────────
MCLR/VPP      ──────     1          Red/Orange
VDD (+3.3V/5V) ──────     2          Red
VSS (Ground)   ──────     3          Black
PGD (Data)     ──────     4          White/Yellow
PGC (Clock)    ──────     5          Brown/Green
```

Important Connection Notes

1. **Power Options:**
    - **PICkit Powers Target**: Connect Pin 2 (VDD) to target VDD
    - **External Power**: Use `-W` option, connect target VDD to external supply

- **Voltage Setting**: Use `-A3.3` or `-A5.0` to set voltage

2. **MCLR Connection:**
    - Always connect Pin 1 to target MCLR/VPP pin
    - Some targets may need pull-up resistor (10kΩ) on MCLR

3. **Ground Connection:**
    - Pin 3 (VSS) must always be connected to target ground
    - This provides the reference for all signals

4. **Programming Pins:**
    - PGD and PGC are the data and clock lines for programming
    - These correspond to specific pins on your target PIC (check datasheet)

Getting Started

Check Connected PICkit

```
# Navigate to pk2cmd directory
cd pk2cmd

# List all connected PICkit devices
./pk2cmd -B. -S

# List devices with firmware versions
./pk2cmd -B. -S#
```

Expected Output: `Unit # Unit type Unit ID OS Firmware 0 PICkit2 2.32.00`

List Supported Device Families

```
# Show all auto-detectable PIC families
./pk2cmd -B. -PF
```

This shows 20 families including Midrange, PIC18F, PIC24, dsPIC33, PIC32, and EEPROMs.

Target Device Operations

Device Detection and Information

```
# Auto-detect connected PIC device (any family)
./pk2cmd -B. -P

# Auto-detect within specific family (e.g., PIC18F family #5)
./pk2cmd -B. -PF5

# Show detailed device information (requires connected target)
./pk2cmd -B. -P -I

# Show device ID and silicon revision for specific part
./pk2cmd -B. -PPIC16F877A -I
```

Power and Voltage Control

```
# Set VDD voltage
./pk2cmd -B. -P -A3.3                # Set to 3.3V
./pk2cmd -B. -P -A5.0                # Set to 5.0V

# Power target after operations (keeps power on)
./pk2cmd -B. -P -T

# Use external power instead of PICkit power
./pk2cmd -B. -P -W -Fprogram.hex -M

# Release MCLR after operations (allows target to run)
./pk2cmd -B. -P -R
```

Programming Operations

Basic Programming

```
# Program hex file to auto-detected device
./pk2cmd -B. -P -Fprogram.hex -M

# Program specific device type
./pk2cmd -B. -PPIC16F877A -Fprogram.hex -M

# Erase device first, then program
./pk2cmd -B. -P -E -Fprogram.hex -M

# Program with verification
./pk2cmd -B. -P -Fprogram.hex -M -Y
```

Selective Memory Programming

```
# Program only program memory
./pk2cmd -B. -P -Fprogram.hex -MP

# Program only EEPROM
./pk2cmd -B. -P -Fprogram.hex -ME

# Program only configuration memory
./pk2cmd -B. -P -Fprogram.hex -MC

# Program only ID memory
./pk2cmd -B. -P -Fprogram.hex -MI

# Program multiple regions
./pk2cmd -B. -P -Fprogram.hex -MPC   # Program + Config
```

Advanced Programming Options

```
# Display programming progress
./pk2cmd -B. -P -Fprogram.hex -M -J

# Set programming speed (1=fastest, 16=slowest)
./pk2cmd -B. -P -L1 -Fprogram.hex -M

# Program with VPP first entry method
./pk2cmd -B. -P -X -Fprogram.hex -M

# Preserve EEPROM data when programming
./pk2cmd -B. -P -Z -Fprogram.hex -M
```

Reading Operations

Read Device to File

```
# Read entire device to hex file
./pk2cmd -B. -P -GFbackup.hex

# Read to specific file path
./pk2cmd -B. -P -GF/path/to/backup.hex
```

Read Memory Regions to Screen

```
# Read program memory range (hex addresses)
./pk2cmd -B. -P -GP0-FF
./pk2cmd -B. -P -GP0-1FF             # Larger range

# Read EEPROM range
./pk2cmd -B. -P -GE0-FF

# Read ID memory
./pk2cmd -B. -P -GI

# Read configuration memory
./pk2cmd -B. -P -GC
```

Device Verification and Testing

Verification Operations

```
# Blank check device
./pk2cmd -B. -P -C

# Verify device against hex file
./pk2cmd -B. -P -Fprogram.hex -Y

# Verify specific memory regions
./pk2cmd -B. -P -Fprogram.hex -YP    # Program memory only
./pk2cmd -B. -P -Fprogram.hex -YE    # EEPROM only
./pk2cmd -B. -P -Fprogram.hex -YC    # Configuration only
```

File Operations

```
# Verify hex file checksum
./pk2cmd -B. -Fprogram.hex -K

# Display hex file information
./pk2cmd -B. -Fprogram.hex -I
```

Device Erase Operations

```
# Erase entire device
./pk2cmd -B. -P -E

# Erase then verify blank
./pk2cmd -B. -P -E -C
```

Typical Programming Workflows

Complete Programming Sequence

```
# 1. Check PICkit connection and firmware
./pk2cmd -B. -S#

# 2. Auto-detect target device
./pk2cmd -B. -P

# 3. Get detailed device information
./pk2cmd -B. -P -I

# 4. Set appropriate voltage
./pk2cmd -B. -P -A5.0

# 5. Erase device
./pk2cmd -B. -P -E

# 6. Blank check
./pk2cmd -B. -P -C

# 7. Program with verification and progress
./pk2cmd -B. -P -Fmyprogram.hex -M -Y -J

# 8. Power target and release MCLR for execution
./pk2cmd -B. -P -T -R
```

Quick Programming (One Command)

```
# Erase, program, verify, power, and release in one command
./pk2cmd -B. -P -E -Fprogram.hex -M -Y -T -R
```

Backup and Restore

```
# Backup device
./pk2cmd -B. -P -GFbackup_$(date +%Y%m%d).hex

# Restore from backup
./pk2cmd -B. -P -E -Fbackup_20231125.hex -M -Y
```

Advanced Features

Multiple PICkit Units

```
# List all connected units
./pk2cmd -B. -S

# Use specific unit by Unit ID (if assigned)
./pk2cmd -B. -SMyPICkit -P

# Assign Unit ID to PICkit
./pk2cmd -B. -NLab1A
```

Programming Speed Control

```
# Fastest programming (default)
./pk2cmd -B. -P -L1 -Fprogram.hex -M

# Slower for problematic connections
./pk2cmd -B. -P -L8 -Fprogram.hex -M

# Slowest for very long cables or noisy environments
./pk2cmd -B. -P -L16 -Fprogram.hex -M
```

Special Device Features

```
# Program OSCCAL value (for devices that support it)
./pk2cmd -B. -P -U3400 -Fprogram.hex -M

# Set I2C address for I2C EEPROMs
./pk2cmd -B. -P -#0x51 -Fdata.hex -M
```

Help and Information Commands

Getting Help

```
# General help
./pk2cmd "-?"

# Help for specific options
./pk2cmd -B. "-?M"                    # Programming help
./pk2cmd -B. "-?G"                    # Reading help
./pk2cmd -B. "-?P"                    # Part selection help
./pk2cmd -B. "-?V"                    # Version information
./pk2cmd -B. "-?E"                    # Exit codes

# List supported devices
./pk2cmd -B. "-?P"

# Search for specific devices
./pk2cmd -B. "-?PPIC16"               # Find PIC16 devices
./pk2cmd -B. "-?PPIC18F"              # Find PIC18F devices
```

Troubleshooting

Common Issues and Solutions

1. **"PK2DeviceFile.dat device file not found"** ```bash

   # Always use -B. for local directory

   ./pk2cmd -B. -S

   # Or specify full path if installed system-wide

   pk2cmd -B/usr/local/share/pk2cmd -S ```

2. **"Auto-Detect: No known part found"**

   - Check all connections (especially VDD, VSS, MCLR, PGD, PGC)
   - Verify target has power (either from PICkit or external)
   - Try slower programming speed: `-L8` or `-L16`
   - Check if target is in reset or has code preventing programming

3. **Programming fails or verification errors**

    - Check VDD voltage setting matches target requirements
    - Verify all ICSP connections are solid
    - Try slower programming speed
    - Check for proper decoupling capacitors on target
    - Ensure MCLR has proper pull-up resistor if required

4. **Target doesn't run after programming**

    - Use `-R` to release MCLR
    - Use `-T` to keep power on
    - Check configuration bits (oscillator settings, etc.)
    - Verify VDD is stable and at correct voltage

Connection Verification

```
# Check PICkit voltages and status
./pk2cmd -B. -P -I

# Test connection with simple operations
./pk2cmd -B. -P           # Should detect device
./pk2cmd -B. -P -GI       # Read ID memory
./pk2cmd -B. -P -C        # Blank check
```

Device-Specific Notes

PIC16F Series

- Typically use 5V programming voltage
- Configuration bits control oscillator and other features
- Some have OSCCAL calibration value

PIC18F Series

- Can use 3.3V or 5V programming
- Larger program memory space
- Enhanced instruction set

PIC24/dsPIC33 Series

- Typically 3.3V devices
- May require Programming Executive (PE) - use `-Q` to disable if needed
- 16-bit architecture

PIC32 Series

- 3.3V devices
- Separate boot flash and program flash regions
- Different programming algorithm

Serial EEPROMs

- Use SPI or I2C interface through PICkit
- Specify correct protocol in device selection
- Some require specific addressing for larger sizes

File Formats

Supported File Types

- **Intel HEX (.hex)**: Standard format for program data
- **Binary (.bin)**: Raw binary data (for serial EEPROMs)

File Handling

```
# Import/Export HEX files
./pk2cmd -B. -P -Fprogram.hex -M      # Import (program)
./pk2cmd -B. -P -GFbackup.hex         # Export (read)

# Binary files (for EEPROMs)
./pk2cmd -B. -P -Fdata.bin -M         # Program binary data
./pk2cmd -B. -P -GFdata.bin           # Read to binary file
```

For build and installation instructions, see BUILD_MACOS.md

For additional help, use: `./pk2cmd "-?"` **or** `./pk2cmd -B. "-?<option>"`